Kuta Software - Infinite Calculus

Name____________________

Exponential Growth and Decay

Date________________ Period____

Solve each exponential growth/decay problem.

1) For a period of time, an island's population grows at a rate proportional to its population. If the growth rate is 3.8% per year and the current population is 1543, what will the population be 5.2 years from now?

2) During the exponential phase, E. coli bacteria in a culture increase in number at a rate proportional to the current population. If the growth rate is 1.9% per minute and the current population is 172.0 million, what will the population be 7.2 minutes from now?

3) Radioactive isotope Carbon-14 decays at a rate proportional to the amount present. If the decay rate is 12.10% per thousand years and the current mass is 135.2 mg, what will the mass be 2.2 thousand years from now?

4) A savings account balance is compounded continuously. If the interest rate is 3.1% per year and the current balance is $1077.00, in how many years will the balance reach $1486.73?

5) A cup of coffee cools at rate proportional to the difference between the constant room temperature of 20.0°C and the temperature of the coffee. If the temperature of the coffee was 86.1°C 3.0 minutes ago and the current temperature of the coffee is 79.9°C, what will the temperature of the coffee be 29.0 minutes from now?

6) During the exponential phase, E. coli bacteria in a culture increase in number at a rate proportional to the current population. If the population doubles in 20.4 minutes, in how many minutes will the population triple?